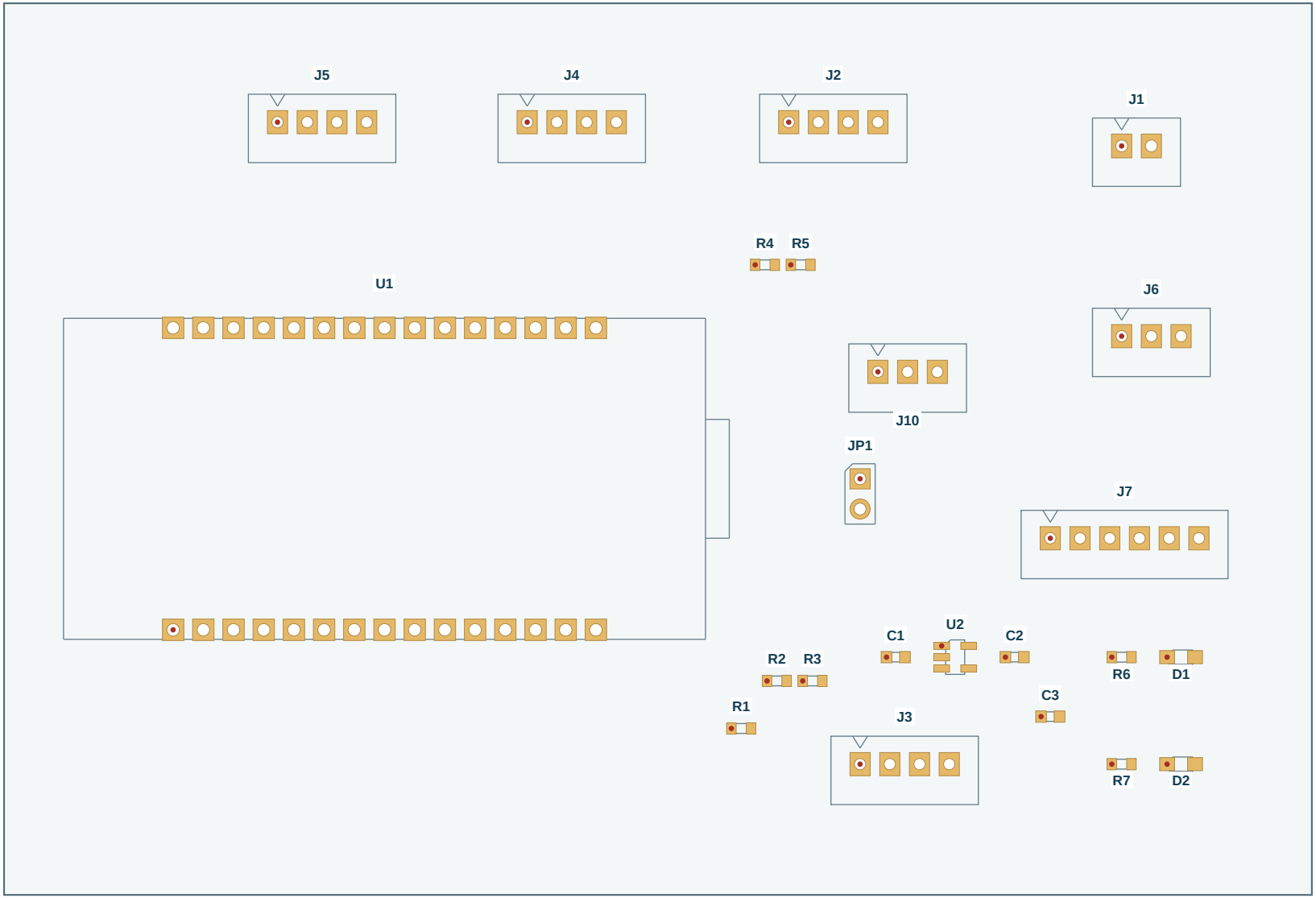
# PCB A - component side and pin 1

ASSEMBLY REVIEW - NOT FOR ORDER | 2026-11-10 | Top view, not mirrored

110 x 75 mm | origin (0,0) = upper-left Edge.Cuts corner




## READING THE DRAWING

Grey: native F.Fab outlines.

Gold: copper lands.

Red dot: native pad 1.

Holes: white centres.

All components on top.

TP pads on BACK: separate map.

Not a stencil/rotation approval.

U1: fit TWO 1x15 female sockets; ESP32 is removable.

Row centres 25.40 mm; contact pitch 2.54 mm. Antenna left, USB right.

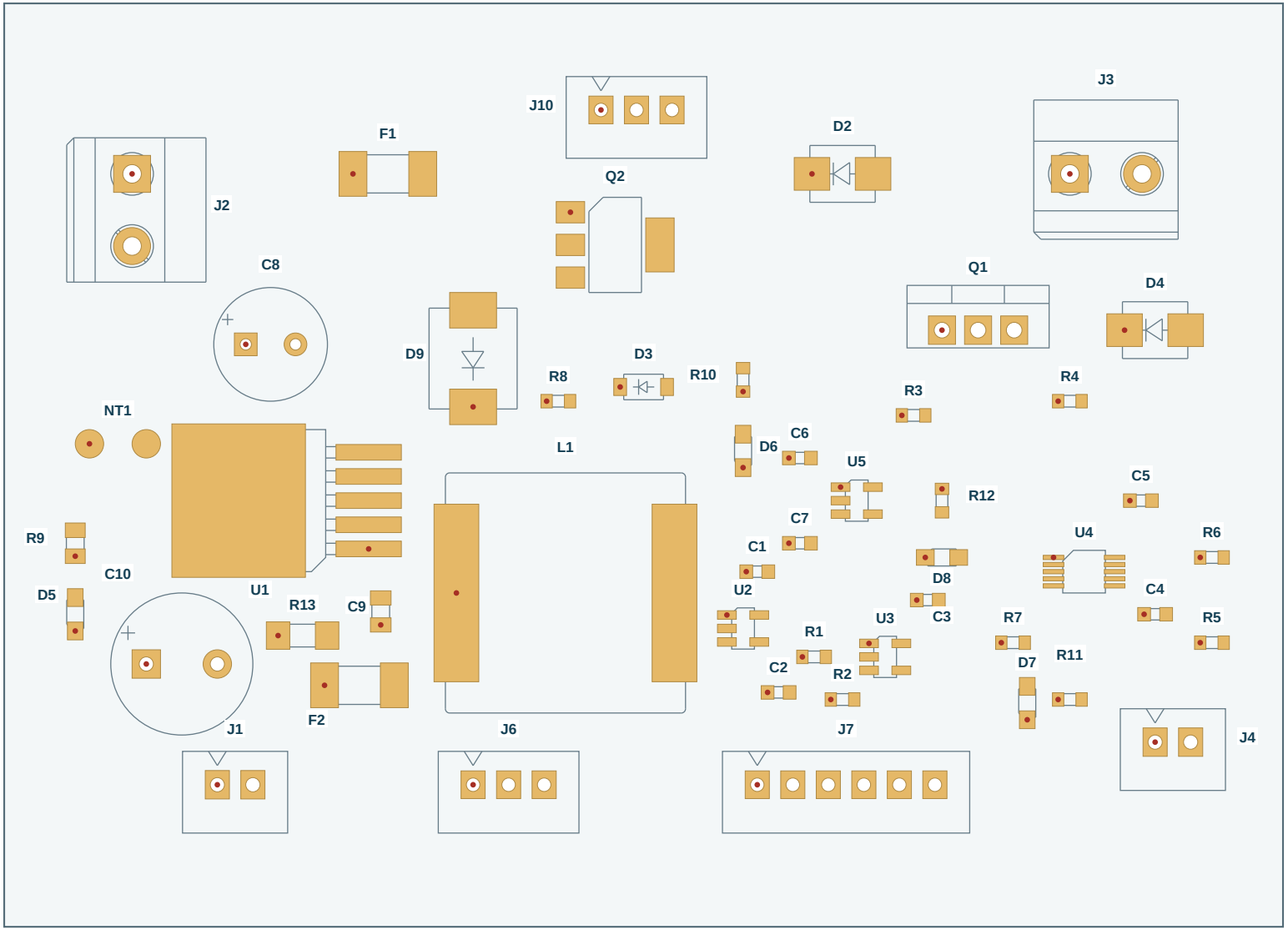
Pads 1..15: bottom row; pads 16..30: top row. Do not mirror.

JP1 shunt supplied loose; follow USB power service instructions.

# PCB B - component side and pin 1

ASSEMBLY REVIEW - NOT FOR ORDER | 2026-11-10 | Top view, not mirrored

90 x 65 mm | origin (0,0) = upper-left Edge.Cuts corner




## READING THE DRAWING

Grey: native F.Fab outlines.

Gold: copper lands.

Red dot: native pad 1.

Holes: white centres.

All components on top.

TP pads on BACK: separate map.

Not a stencil/rotation approval.

J2 = 24 V input; J3 = heater; J4 = passive NTC probe (15k).
C8/C10: pad 1 positive. D9: pad 1 cathode, BUCK_SW.
U1 tab/pad 3 = PGND. NT1 = copper only; do not populate.
R13 uses Vishay-specific lands; see page 3.

# Assembly details - selected orientations and R13 lands

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## R13 / WSL1206R0500FEA - manufacturer recommended pattern, 0.006..0.200 ohm range

Lands: 1.65 x 1.93 mm; centre distance: 3.45 mm

Inner gap: 1.80 mm; total copper span: 5.10 mm

Vishay datasheet 30100, page 2; not a generic 1206 replacement rule.

## Native pad functions to check in the assembler placement preview

| | |
|---|---|
| A U2 / B U2 | AP2112: 1=VIN, 2=GND, 3=EN, 4=NC, 5=3V3 |
| A D1,D2 / B D5..D8 | LED pad 1 = cathode; pad 2 = anode |
| B U1 | LM2596 fixed5V: 1=VIN, 2=SW, 3/tab=PGND, 4=FB(5V), 5=PGND |
| B U3 | 1=PWM0, 2=ENABLE, 3=GND, 4=GATE_CMD, 5=+3V3_B |
| B U5 | 1=+5V_AB, 2=PGND, 3=GATE_CMD, 4=PGND, 5=GATE_DRV |
| B Q1 / Q2 | Q1 G-D-S = 1-2-3; Q2 G-D-S = 1-2-3 (tab=2) |
| B D3 / D4 / D9 | Pad1 cathode: VIN_PROTECTED / VIN_PROTECTED / BUCK_SW respectively |
| B C8 / C10 | Positive pin1: VIN_PROTECTED / +5V_BUCK; C10 pin2 returns via R13 |
| A/B J1; J6; J7; J10 | Pin1 GND. J10:2=TX_A,3=RX_A; pins2/3 NC on temperature B. |
| B J2 / J3 / J4 | Pin1: VIN_RAW / VIN_PROTECTED / 3V3_B. J4 pin2=NTC_RAW. |
| A U1 sockets | Bottom pads1..15 EN->VIN; top pads16..30 3V3->GPIO23; pitch2.54 |
| B NT1 | No purchased or fitted part; PGND/GND copper junction |

Full pin/net/coordinate map: assembly_pad_map.csv. Native origin differs from board-corner origin.

Read these functional notes with the schematics; assembly approval does not validate power or thermal behavior.